

23009200111 陆文萱

1. 下列数列当 $n \rightarrow \infty$ 时是否收敛? 如果收敛, 求出极限.

$$(1) \alpha_n = \frac{1+ni}{1-ni} = \frac{1-n^2+2ni}{1+n^2}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1-n^2}{1+n^2} = -1 \quad \lim_{n \rightarrow \infty} b_n = \frac{2n}{1+n^2} = 0$$

收敛. $\lim_{n \rightarrow \infty} \alpha_n = -1$

$$(2) \alpha_n = (1 + \frac{i}{2})^{-n} \quad [\text{用指数表示法}]$$

$$r = \sqrt{1 + (\frac{1}{2})^2} = \frac{\sqrt{5}}{2} \quad \theta = \arctan \frac{1}{2}$$

$$\therefore \alpha_n = (\frac{\sqrt{5}}{2} e^{i\theta})^{-n} = (\frac{\sqrt{5}}{2})^{-n} e^{-in\theta} = (\frac{2}{\sqrt{5}})^n (\cos n\theta - i \sin n\theta)$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (\frac{\sqrt{5}}{2})^{-n} \cos n\theta = 0$$

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} -(\frac{2}{\sqrt{5}})^n \sin n\theta = 0$$

$$\therefore \lim_{n \rightarrow \infty} \alpha_n = 0$$

$$(3) \alpha_n = e^{-i\frac{n\pi}{2}} = \cos(-\frac{n\pi}{2}) + i \sin(-\frac{n\pi}{2}) \\ = -i \sin \frac{n\pi}{2}$$

当 $n = 4k+1$ 时, $k \in \mathbb{N}$, $\lim_{n \rightarrow \infty} b_n = -1$, 当 $n = 4k+3$ 时, $\lim_{n \rightarrow \infty} b_n = 1$

$n = 4k+2$, 或 $4k+4$, $\lim_{n \rightarrow \infty} b_n = 0$

$\therefore \{\alpha_n\}$ 发散.

2. 判断下列级数收敛性.

$$(1) \sum_{n=1}^{\infty} \frac{i^n}{n}$$

$$\frac{i^n}{n} = \frac{1}{n} e^{i\frac{n\pi}{2}} = \frac{1}{n} (\cos \frac{n\pi}{2} + i \sin \frac{n\pi}{2})$$

$$\sum_{n=1}^{\infty} a_n = \sum_{k=1}^{\infty} \frac{1}{2k} \cdot (-1)^k$$

$$\sum_{n=1}^{\infty} b_n = \sum_{k=1}^{\infty} \frac{1}{(2k-1)} (-1)^{k+1}$$

均为收敛的交错级数

$\therefore$ 级数收敛. 又 $\sum_{n=1}^{\infty} |\frac{i^n}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ 发散, 故原级数条件收敛

$$(2) \sum_{n=2}^{\infty} \frac{i^n}{\ln n}$$

$$\frac{i^n}{\ln n} = \frac{1}{\ln n} (\cos \frac{n\pi}{2} + i \sin \frac{n\pi}{2})$$

$$\therefore \sum_{n=2}^{\infty} a_n = \sum_{k=1}^{\infty} \frac{1}{\ln(2k)} \cdot (-1)^k \quad \sum_{n=2}^{\infty} b_n = \sum_{k=2}^{\infty} \frac{1}{\ln(2k-1)} (-1)^{k+1}$$

均为收敛的交错级数

$\therefore$ 级数收敛. 又 $\sum_{k=2}^{\infty} |\frac{i^n}{\ln n}| = \sum_{n=2}^{\infty} \frac{1}{\ln n} > \sum_{n=2}^{\infty} \frac{1}{n}$, 由比较判别法知原级数条件收敛.

$$(3) \sum_{n=0}^{\infty} \frac{\cos(in)}{2^n}$$

$$\text{注意 } \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\frac{\cos(in)}{2^n} = \frac{e^n + e^{-n}}{2^{n+1}}$$

$$\therefore \sum_{n=0}^{\infty} \frac{e^n}{2^{n+1}} \text{ 发散, } \sum_{n=0}^{\infty} \frac{1}{e^n 2^{n+1}} \text{ 收敛}$$

$\therefore$ 原级数发散.

3. 求下列幂级数收敛半径.

$$(1) \sum_{n=1}^{\infty} \frac{z^n}{n^p} \quad (p \text{ 为正整数})$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{n^p}} = 1$$

$$\therefore R = 1$$

$$(2) \sum_{n=1}^{\infty} \frac{(n!)^2}{n^n} z^n$$

$$\lim_{n \rightarrow \infty} \frac{C_{n+1}}{C_n} = \lim_{n \rightarrow \infty} \frac{[(n+1)!]^2}{(n+1)^{n+1}} \cdot \frac{n^n}{(n!)^2} = \lim_{n \rightarrow \infty} \frac{n+1}{e} = \infty$$

$$\therefore R = 0$$

$$(3) \sum_{n=1}^{\infty} \left(\frac{z}{\ln(in)} \right)^n$$

$$\ln(in) = \ln n + i \cdot \frac{\pi}{2}$$

$$C_n = \left[\frac{1}{\ln(in)} \right]^n$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| = \lim_{n \rightarrow \infty} \frac{|\ln n + i \cdot \frac{\pi}{2}|^{n+1}}{|\ln(n+1) + i \cdot \frac{\pi}{2}|^{n+1}} = 0$$

$$\therefore \text{收敛半径为 } \infty$$

4. 求下列函数在 z_0 处的 Taylor 展开式, 并求收敛半径.

$$(1) \frac{z-1}{z+1}, z_0 = 1.$$

$$\text{奇点 } z = -1, d = |z_0 - z| = 2, \quad |z-1| < 2, \quad \left| \frac{z-1}{2} \right| < 1$$

$$\frac{z-1}{z+1} = \frac{z-1}{z-1+2} = \frac{\frac{z-1}{2}}{\frac{z-1}{2}+1} = 1 - \frac{1}{1 - (-\frac{z-1}{2})} = 1 - \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-1}{2} \right)^n$$

$$= \frac{z-1}{2} - \left(\frac{z-1}{2} \right)^2 + \left(\frac{z-1}{2} \right)^3 - \dots + (-1)^{n+1} \left(\frac{z-1}{2} \right)^n$$

$$R = 2$$

$$(2) \frac{z}{(z+1)(z+2)}, z_0 = 2$$

最远距离

$$\text{奇点 } z = -1, -2, d = |-1 - z_0| = 3, \quad |z-2| < 3, \quad \left| \frac{z-2}{3} \right| < 1$$

$$\frac{z}{(z+1)(z+2)} = \frac{z}{z+2} - \frac{1}{z+1}$$

$$= \frac{1}{2} \cdot \frac{1}{1 + \frac{z-2}{4}} - \frac{1}{3} \cdot \frac{1}{1 + \frac{z-2}{3}} \quad \left| \frac{z-2}{4} \right| < 1, \quad \left| \frac{z-2}{3} \right| < 1$$

$$= \frac{1}{2} \cdot \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{4} \right)^n - \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-2}{3} \right)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n \cdot (z-2)^n \left[\frac{1}{2 \cdot 4^n} - \frac{1}{3^{n+1}} \right] \quad \text{收敛半径为 } 3$$

$$(3) \frac{1}{z^2}, z_0 = -1,$$

$$\text{奇点 } z = 0, d = |z_0 - z| = 1, \quad |z+1| < 1,$$

$$\frac{1}{z^2} = -\left(\frac{1}{z} \right)' \quad [\text{逐项求导}]$$

$$\frac{1}{z} = + \frac{1}{1+z-1} = + \sum_{n=0}^{\infty} (-1)^n (z-1)^n = + (1 - (z-1) + (z-1)^2 + \dots)$$

$$\therefore -\left(\frac{1}{z} \right)' = -(-1 + 2(z-1) - 3(z-1)^2 + \dots)$$

$$= 1 - 2(z-1) + 3(z-1)^2 + \dots$$

$$= \sum_{n=1}^{\infty} (-1)^{n-1} \cdot n \cdot (z-1)^{n-1} \quad \text{收敛半径为 } 1$$

$$(4) \arctan z, z_0 = 0$$

无奇点

[逐项积分]

$$(\arctan z)' = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n \cdot z^{2n}, \quad |z^2| < 1, \quad |z| < 1$$

$$\therefore \arctan z = \int_0^z \sum_{n=0}^{\infty} (-1)^n z^{2n} dz = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{2n+1}, \quad \text{收敛半径为 } 1$$

5. 求下列函数在给定圆环域内展开成 Laurent 级数

(1) $\frac{\ln(2-z)}{z(z-1)}$, $0 < |z-1| < 1$.

奇点 0, 1, 2.

$$\ln(1+z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{n+1}}{n+1}$$

$$\ln(2-z) = \ln[1-(z-1)] = \sum_{n=0}^{\infty} (-1)^n \frac{[-(z-1)]^{n+1}}{n+1} = \sum_{n=0}^{\infty} -\frac{(z-1)^{n+1}}{n+1}$$

$$\therefore \frac{\ln(2-z)}{z-1} = \sum_{n=0}^{\infty} -\frac{(z-1)^n}{n+1}$$

$$\frac{1}{z} = \frac{1}{1+(z-1)} = \sum_{n=0}^{\infty} (-1)^n \cdot (z-1)^n$$

$$\therefore \frac{\ln(2-z)}{z-1} \cdot \frac{1}{z} = - \sum_{n=0}^{\infty} \frac{(z-1)^n}{n+1} \cdot \sum_{n=0}^{\infty} (-1)^n \cdot (z-1)^n \quad ?$$

$$C_n = \sum_{k=0}^n a_k b_{n-k}$$

$$= \sum_{k=0}^n (-1)^k (-1) \cdot \frac{1}{n-k+1}$$

$$= -\left(\frac{1}{n-1} - \frac{1}{n-2} + \frac{1}{n-3} \dots\right)$$

(2) $e^{\frac{1}{1-z}}$, $1 < |z| < +\infty$?

$z=1$ 是奇点

$$\therefore 1 < |z| < +\infty$$

$$\therefore \left|\frac{1}{z}\right| < 1$$

$$\therefore \frac{1}{1-z} = \frac{1}{z(\frac{1}{z}-1)} = -\frac{1}{z(1-\frac{1}{z})} = -\frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right) = -\left(\frac{1}{z} + \frac{1}{z^2} + \dots\right)$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} \left(\sum_{k=0}^{\infty} z^{-k}\right)^n \cdot (-1)^n$$

$$\therefore \text{原式} = 1 - \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) + \frac{1}{2!} \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right)^2 - \frac{1}{3!} \left(\frac{1}{z} + \frac{1}{z^2} + \dots\right)^3 + \dots$$

(3) $\sin \frac{1}{1-z}$ 在 $z=1$ 的去心邻域内 $= \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$

$$\therefore \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots, \quad |z| < +\infty$$

$$\therefore \sin \frac{1}{1-z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot \frac{1}{(1-z)^{2n+1}}$$

(4) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, $3 < |z| < 4$; $4 < |z| < +\infty$

(i) $3 < |z| < 4$

$$\text{原式} = 1 - \frac{2}{z-3} + \frac{6}{z-4} \quad \left|\frac{3}{z}\right| < 1, \quad \left|\frac{4}{z}\right| < 1$$

$$= 1 - \frac{3}{z} \cdot \frac{1}{1-\frac{3}{z}} - \frac{2}{z} \cdot \frac{1}{1-\frac{4}{z}}$$

$$= 1 - \frac{3}{z} \left(1 + \left(\frac{3}{z}\right)^1 + \left(\frac{3}{z}\right)^2 + \dots\right) - \frac{2}{z} \left(1 + \left(\frac{4}{z}\right)^1 + \left(\frac{4}{z}\right)^2 + \dots\right)$$

$$= 1 - \frac{3}{z} \sum_{n=0}^{\infty} \left(\frac{3}{z}\right)^n - 2 \sum_{n=0}^{\infty} \frac{4^n}{z^{n+1}}$$

6. 求下列函数奇点的类型, 求极点阶数, 对无穷远点讨论.

(1) $\frac{z-1}{z(z^2+4)^2}$ 分子、分母无公共零点

令 $z(z^2+4)^2=0$, $z=0, \pm 2i$, 分别是分母的 1 级和 2 级零点,

$\therefore 0, \pm 2i$ 分别是函数的 1 级和 2 级极点.

$$\therefore \text{当 } z \rightarrow \infty, \quad \frac{z-1}{z(z^2+4)^2} \rightarrow 0$$

$\therefore z = \infty$ 是可去奇点

(2) $\frac{1}{e^z - 1}$

令 $e^z - 1 = 0$, $z = 2k\pi i$, $k \in \mathbb{Z}$

$\therefore (e^z - 1)'|_{z=2k\pi i} = e^{2k\pi i} = \cos 2k\pi + i \sin 2k\pi = 1 \neq 0$

$\therefore z = 2k\pi i$ 为分母的一阶零点, 而分子是常数 1

$\therefore 2k\pi i$ 为函数的一阶极点.

$\therefore$ 令 $k \rightarrow \infty$, $z \rightarrow \infty$, $z = \infty$ 为极点的聚点

$\therefore \infty$ 非孤立奇点

(3) $\frac{1 - \cos z}{z^3}$

令 $z^3 = 0$, $z = 0$ 为奇点.

$\therefore \lim_{z \rightarrow 0} \frac{1 - \cos z}{z^3} = \lim_{z \rightarrow 0} \frac{\sin z}{3z^2} = \infty$

$\therefore$ 在 ∞ 的去心邻域, 函数的 Laurent 展开式为

$$\frac{1}{z^3} [1 - (1 - \frac{1}{2}z^2 + \frac{1}{24}z^4 - \dots)] = \frac{1}{2z} - \frac{1}{24}z + \frac{1}{6!}z^3 \dots$$

$\therefore 0$ 为 $f(z)$ 的 1 级极点

$\therefore$ 含无穷多正幂项

$\therefore \infty$ 是本性奇点 (极限不存在且不为 ∞)

(4) $\sin \frac{1}{\sin \frac{1}{z}}$

$\sin \frac{1}{z} = 0$, $z = \frac{1}{k\pi}$, $k \in \mathbb{Z}$ 且 $k \neq 0$.

$z = 0$,

$\therefore z = 0$ 为奇点但不是孤立奇点 $\lim_{k \rightarrow \infty} \frac{1}{k\pi} = 0$.

$z = \frac{1}{k\pi}$ 为孤立奇点

$z \rightarrow \frac{1}{k\pi}$, $\lim_{z \rightarrow \frac{1}{k\pi}} f(z)$ 不存在且不为 ∞

$\therefore z = \frac{1}{k\pi}$ 为本性奇点